- Stub axle is capable of limited angular movement about the kingpin for steering the vehicle.
- Stub axle helps to turn the vehicle at desired direction through steering linkage.

8.12.1 Types of Stub axle

Front axles are built according to four basic designs for attaching the steering knuckle to the beam. They are

1. Elliot type
2. Reverse Elliot type
3. Lemoine type
4. Reverse Lemoine type

1. Elliot axle

The Elliot type stub axle is shown in the figure 8.12.1 (a). The end of the axles are in the shape of a elliot. A king pin with cotter pin connects the stub axle. So the stub axle turns easily to and fro.

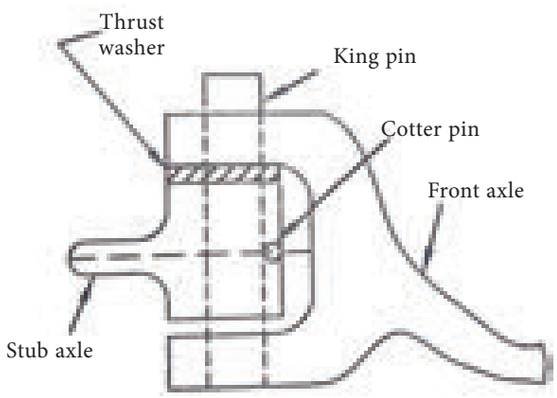


Figure 8.12.1 (a) Elliot type stub axle

2. Reversed Elliot axle

The reverse elliot type is shown in figure 8.12.1 (b) of the stub axle in the reversed Elliot shape. A king pin with cotter pin connects the stub axle. So the stub axle turns easily to and fro.

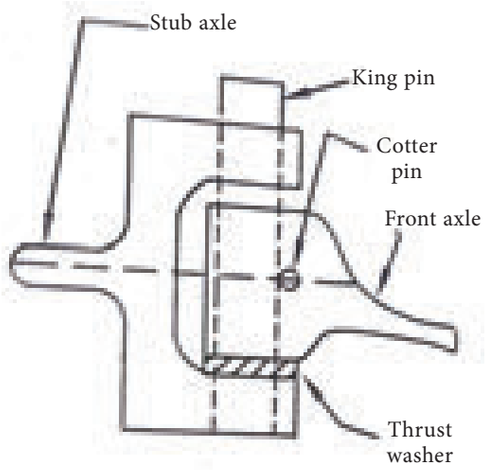


Figure 8.12.1 (b) Reversed elliot

3. Lemoine axle

Two spindles at right angles to each other, stub axle and axle center. Lemoine, a French axle and spring manufacturer designed this. In this design, pivot spindle extended up from the wheel spindle. Tends to raise the axle center above the wheel spindle, hence poorly suited to modern conditions, where low axle centres is preferred.

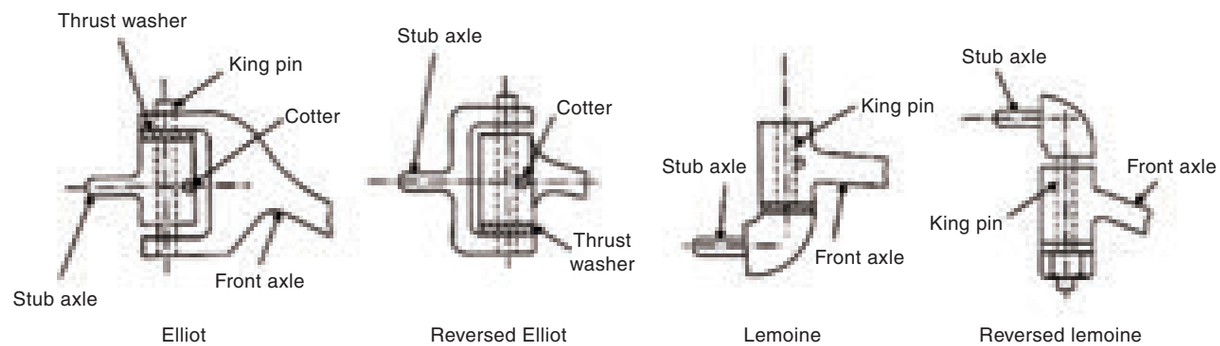


Figure 8.12.1 (a)

BAE_ENG_Chapter_08.indd 161 12/10/2019 12:11:21 PM